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Introduction

The hydrocarbons are compounds composed entirely of Carbon and Hydrogen. They occur widely in nature. The crude oil consists of complex mixtures of hydrocarbons, mangoes contain cyclohexane, a cyclic hydrocarbon, cockroaches secretes a hydrocarbon, undecane which attract opposite gender of its species. Hydrocarbons are primarily used as fuel. For example, Liquefied mixture of propane and butane is used as Liquefied petrolium gas (LPG). They also finds many applications in industries such as solvents etc. In this unit we will study the classification, preparation, properties and uses of aliphatic and aromatic hydrocarbons.

13.1 Introduction and classification of alkanes:

Depending upon the characteristic pattern of bonding between the carbon atoms, hydrocarbons are divided into two main classes: aliphatic and aromatic. The word aliphatic was derived from the Greek word 'aleiphar' meanings fat. Important sources of aliphatic hydrocarbons are oils and fats. The word 'aroma' means odour, which is obtained by chemical treatment of pleasant-smelling plant extracts.

Aliphatic hydrocarbons include three major groups: alkanes, alkenes and alkynes. Alkanes are saturated hydrocarbons in which all the bonds between the carbon atoms are single bond, alkenes consist of atleast one carbon-carbon double bond, and alkynes have atleast one carbon-carbon triple bond. Hydrocarbons having localised carbon - carbon multiple bonds are called unsaturated hydrocarbons.

Aromatic hydrocarbons are cyclic compounds which contain characteristic benzene ring or its derivatives. The classification of hydrocarbons is as shown below .

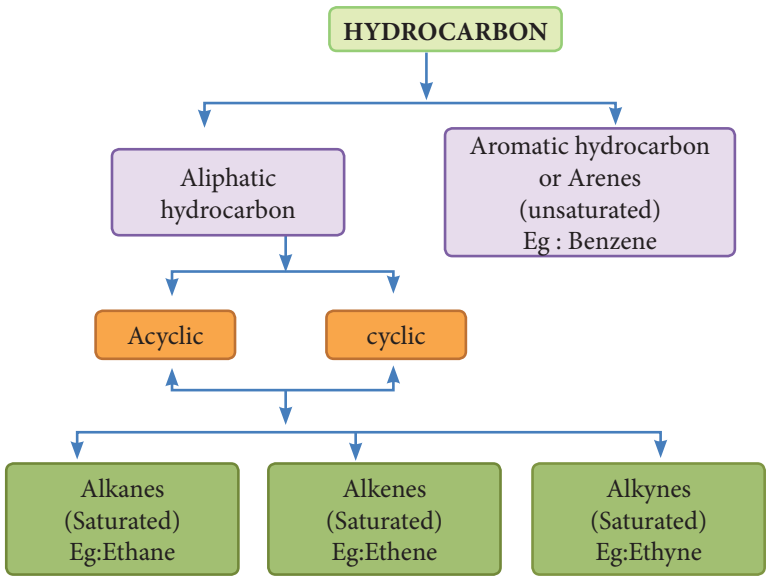

Fig.13.1. Classification of Hydrocarbons